Team Seaborn

OPTIMIZING CUSTOMER CONVERSION RATES IN E-COMMERCE

A Data-Driven Analysis of Customer Behavior and Factors Influencing Conversion

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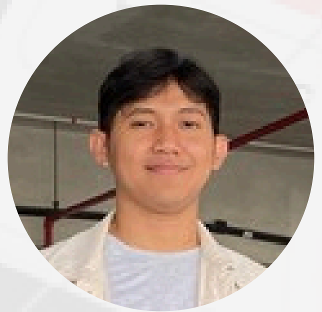

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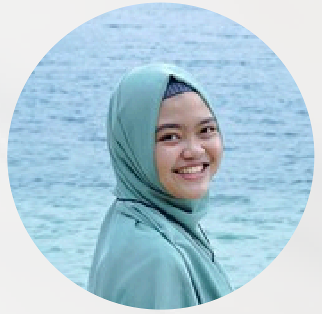


Roya

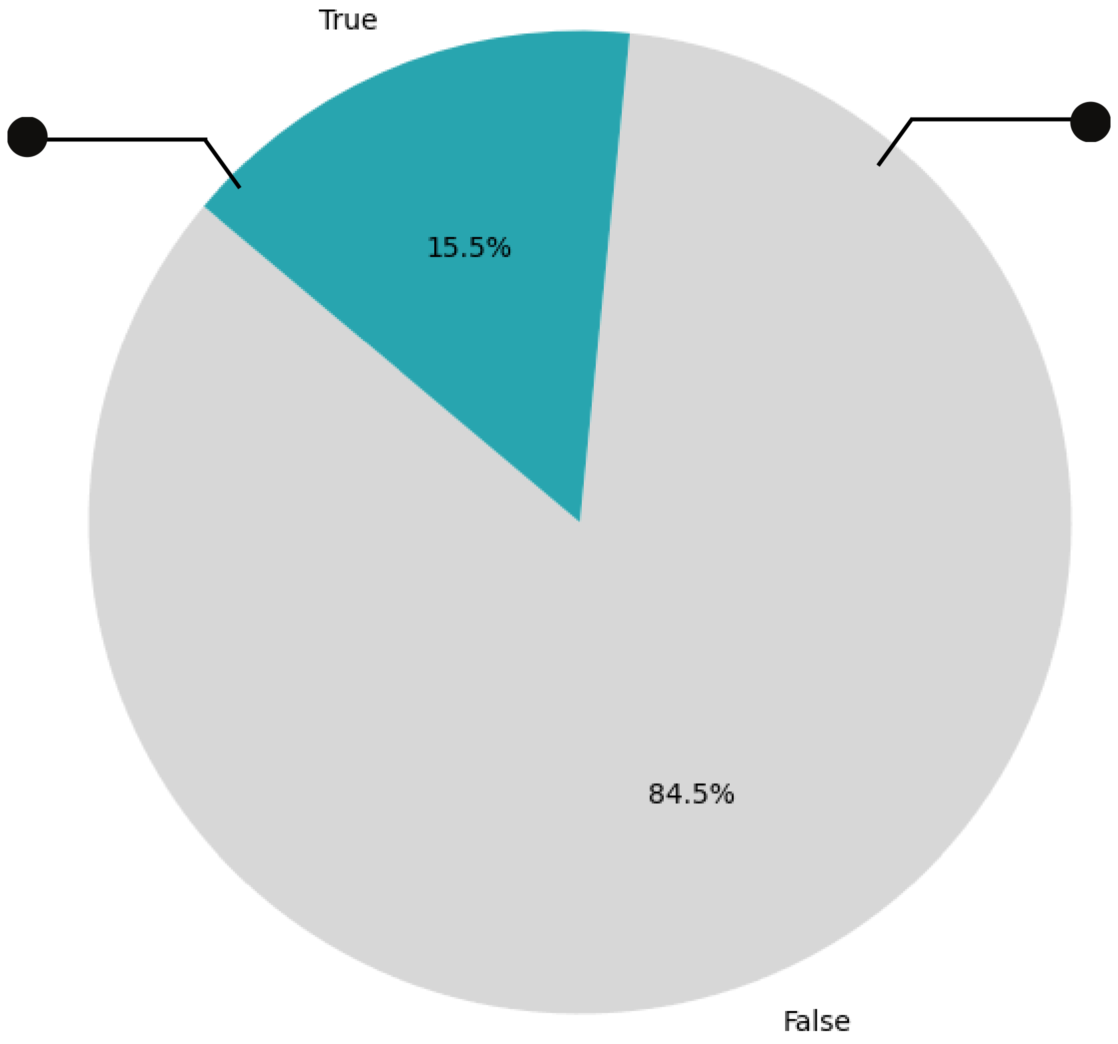
Content

Problem

ShopEase is a leading e-commerce platform that provides a wide range of products to customers.

Background

15.5% of customers were successfully converted from all visitors to the company's website. The **average monthly** customer conversion rate was **14.45%.**



Problem

Maintaining and **improving conversion rates** remains a challenge amid fierce competitors and changing consumer behavior.

Unfortunately, the company **still lacks** a deep understanding of the **factors** that **influence customer conversion.**

Urgency

Without an understanding of conversion factors and predictive tools, **companies risk missing opportunities to optimize their marketing strategies** and **face potential declines** in customer conversion.

Goal

01

Identifying factors that influence customer conversion

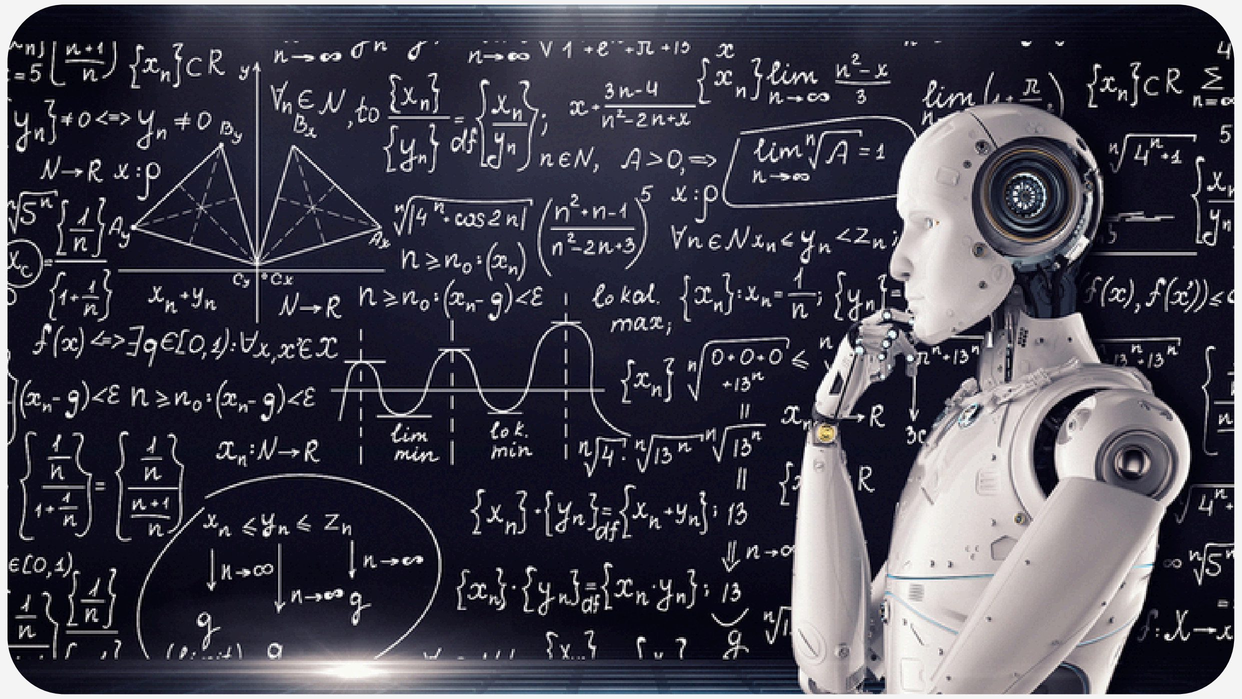
02

Increasing company sales

03

Anticipating a further decline in sales

Method



Machine Learning (ML)

ML is a branch of AI that enables systems to learn from data, identify patterns, and make decisions with minimal human intervention. Systems can learn with little or no supervision.

Why Machine Learning (ML) is so popular in the 21st Century

Advantages of ML

- 61% of business decision-making emphasizes the application of AutoML
- ML is revolutionizing the marketing sector, where it can increase and improve customer satisfaction by more than 10%.
- According to McKinsey, companies that adopt AL and ML have seen productivity increases of up to 40%.
- The Boston Consulting Group reports that companies using AI can reduce operating costs by 20-30%.

About Dataset

E-Commerce Dataset

Sumber: Sakar, C.O., Polat, S.O., Katircioglu, M. et al. Neural Comput & Applic (2018).

Consisting of 12,946 observations

It has 18 columns, 10 numeric columns and 8 categorical columns.

Consisting of columns:

1. Administrative
2. Administrative Duration
3. Informational
4. Informational Duration
5. Product Related
6. Product Related Duration
7. Bounce Rates
8. Exit Rates
9. Page Value
10. Special Day
11. Month
12. Operating Systems
13. Browser
14. Region
15. Traffic Type
16. Visitor Type
17. Weekend
18. Revenue (label)

Business Question

- What are the company's conversion trends?
- What factors influence visitors to generate revenue (conversion)?
- How can ML improve the company's conversion rate?
- What can be done to improve the company's conversion rate?

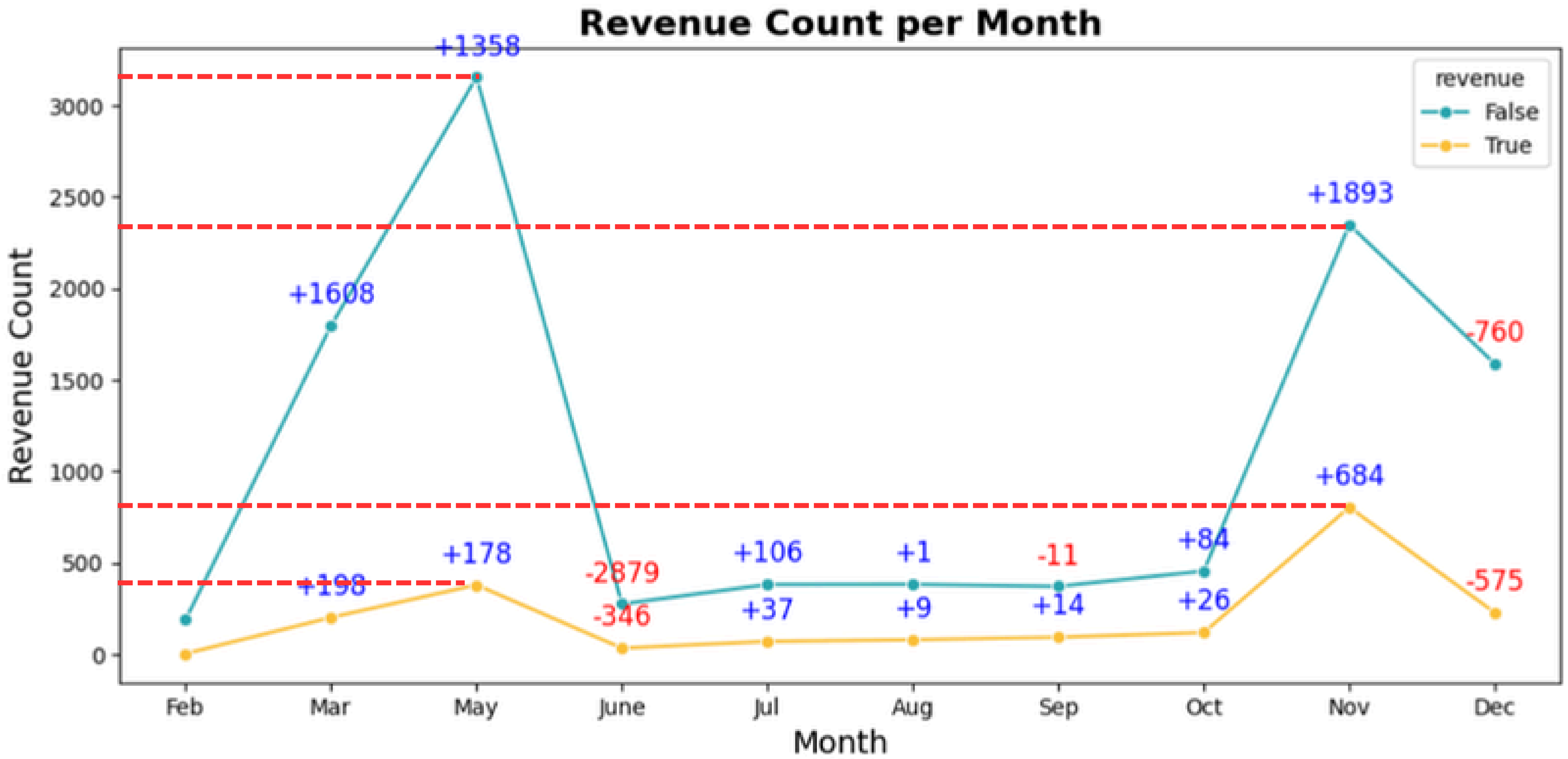
Business Metric

The business metric used is **Conversion Rate** (the percentage of visitors who generate revenue from all visitors on the company's platform).

Explanatory Data Analysis

BI-VARIATE ANALYSIS

INCREASE IN THE NUMBER OF VISITORS IN MEI AND NOVEMBER



Revenue Count :

May

True : +178 visitor

False : +1.358 visitor

November

True : +684 visitor

False : +1.893 visitor

The company still struggles to optimize its potential (high visitor numbers) and lacks understanding of conversion factors.

BI-VARIATE ANALYSIS

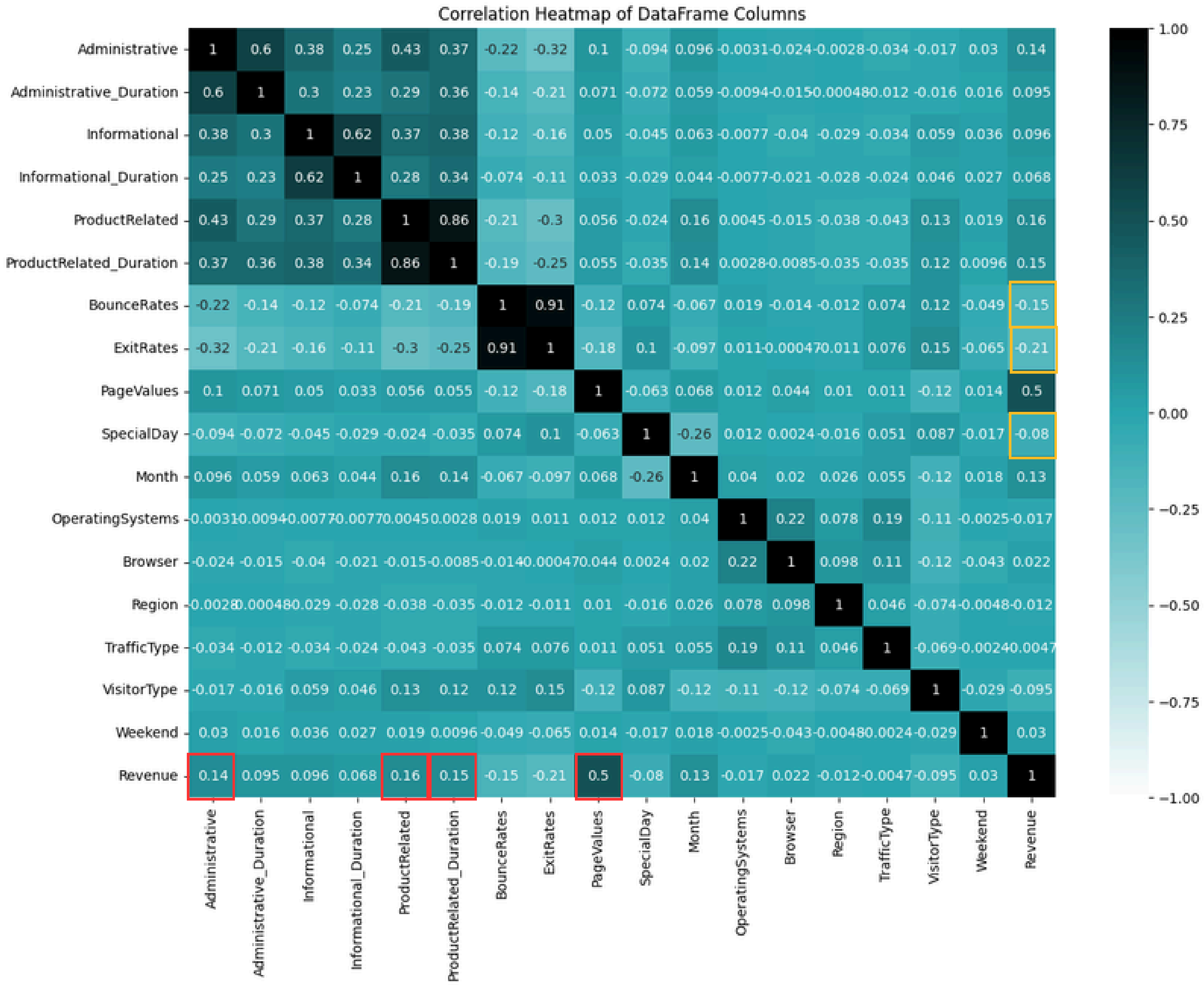
REVENUE COUNT BY VISITOR TYPE PER MONTH

Returning visitors dominated over others,
but **the gap** with **new visitors** was quite good.

It is recommended to focus on improving the system and methods so that it can convert returning visitors.

BI-VARIATE ANALYSIS

CORRELATION HEATMAP OF DATAFRAME COLUMNS



Correlation Revenue between Features:

Positive (the higher the value = the more likely to convert)

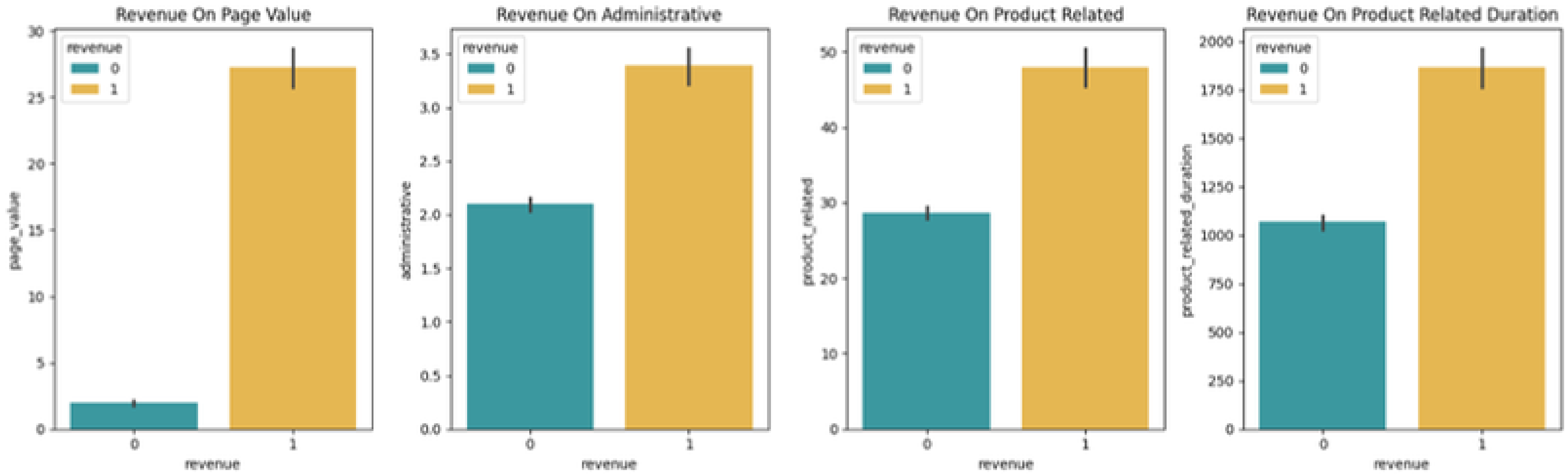
- Page Values: 0,5
- Product Related: 0,16
- Product Related Duration: 0,15
- Administrative: 0,14

Negative (inverse relationship with revenue)

- Bounce Rates: -0,15
- Exit Rates: -0,21
- Special Day: -0,08

BI-VARIATE ANALYSIS

REVENUE BY CORRELATION FEATURE POSITIVE

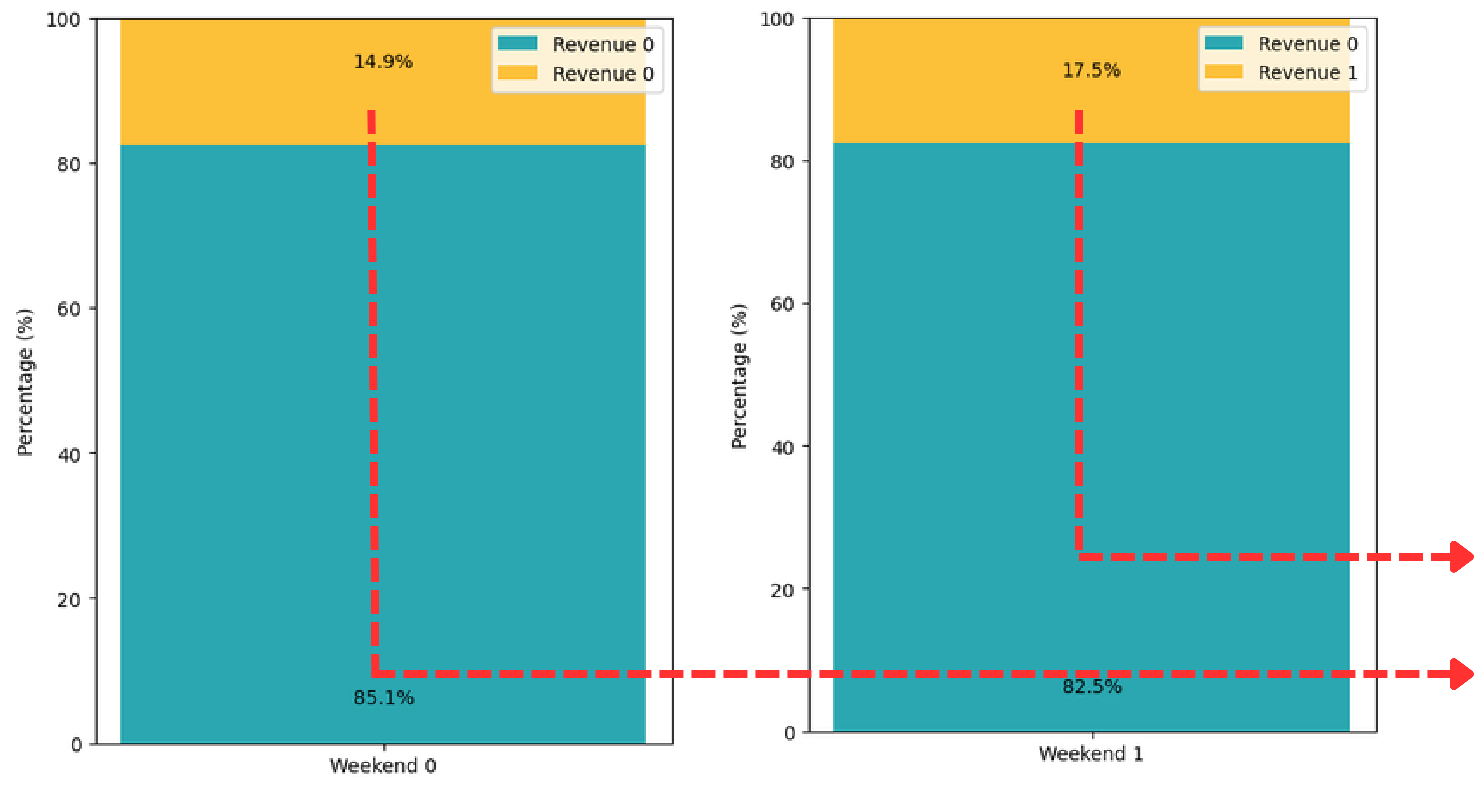


Customers who exhibit higher Page Value, Administrative activity, Product Related interactions, and Product Related Duration **tend to have a greater propensity to convert**.

Focus on relevant content to convert visitors into loyal customers.

BI-VARIATE ANALYSIS

REVENUE BY WEEKDAYS, WEEKEND & SPECIAL DAY



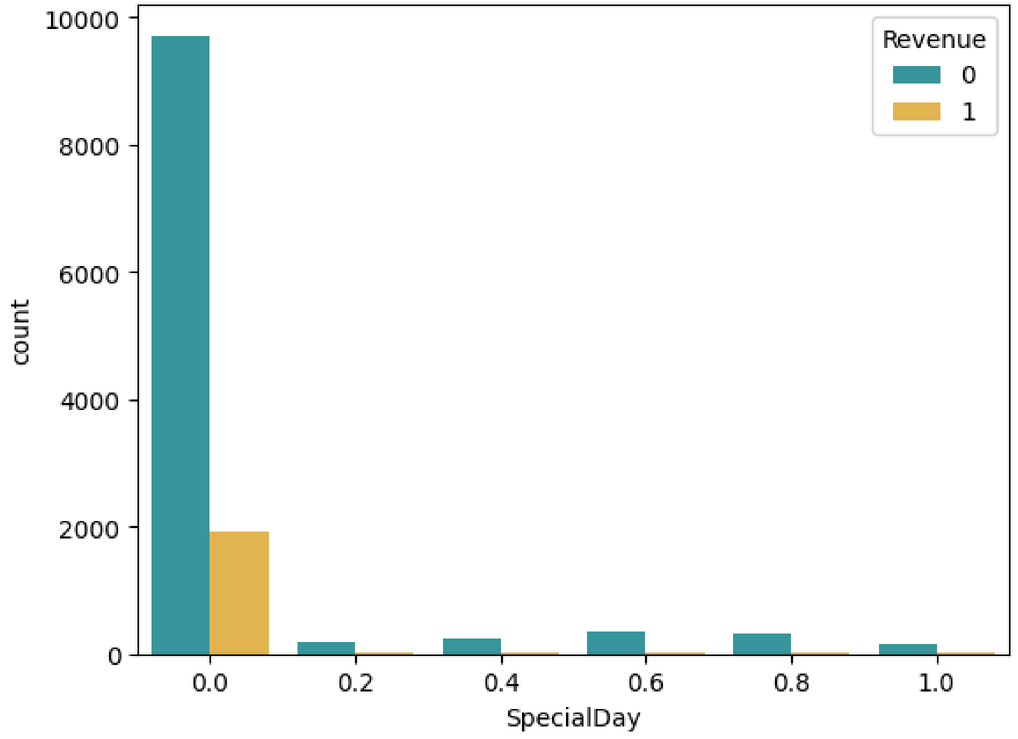
MANY CONVERTED VISITORS ON THE WEEKEND

Revenue Conversion :

Weekend : 17,5%

Weekday : 14,9%

Special promotions and **attractive offers** are **prioritized** on **weekends.**



Special Days have no effect on conversion rates.

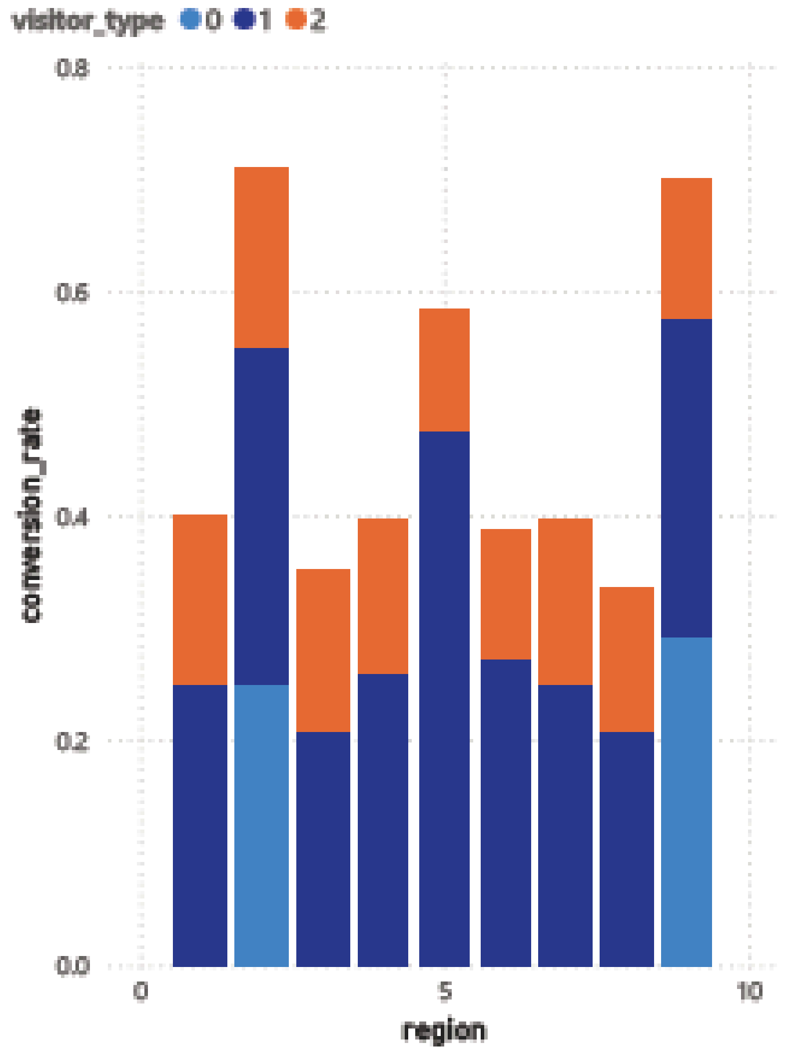
Improve the platform and **enhance** the user experience

Dashboard

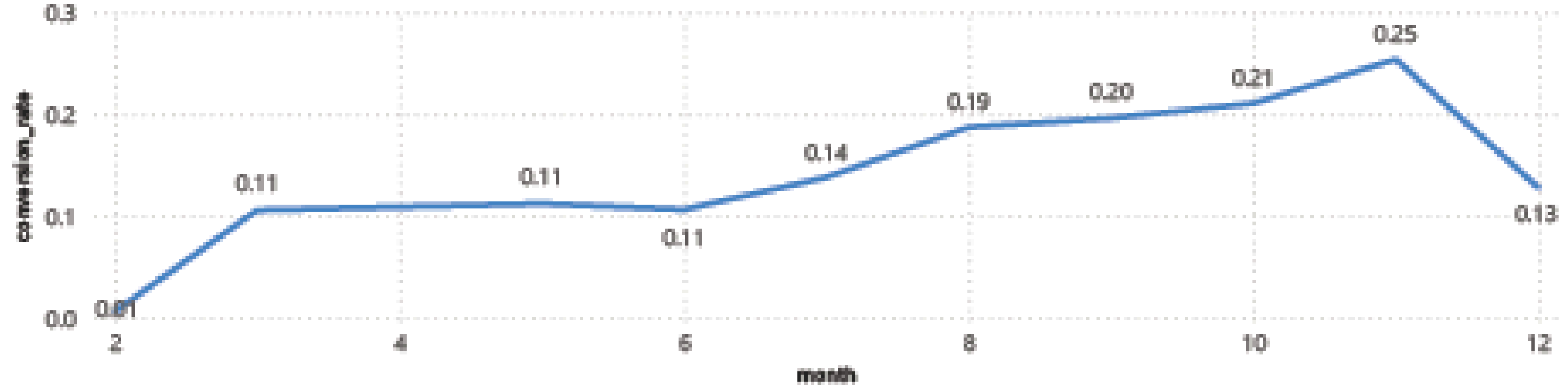
OVERVIEW PROBLEM

0.16	10K	1648	702.31M	7.15M
conversion_rate	total_sessions	total_conversions	avg_page_value	avg_bounce_rate

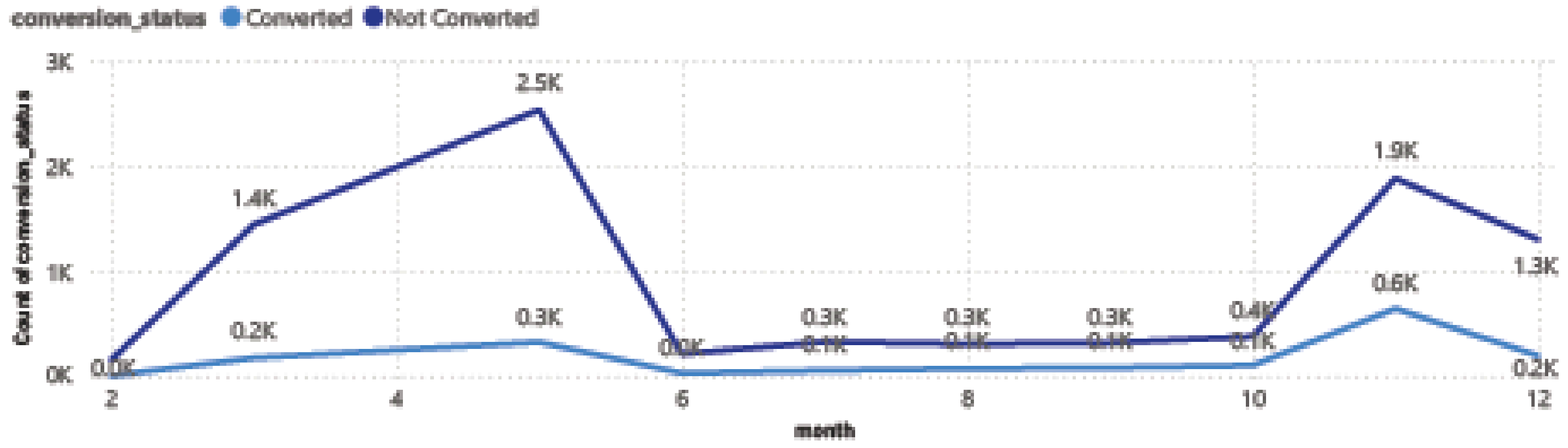
Conversion Rate for Each Region



Conversion Rate by Month



Conversion Status by Month



BEHAVIOR CONCERN

0.16
conversion_rate

10K
total_sessions

1648
total_conversions

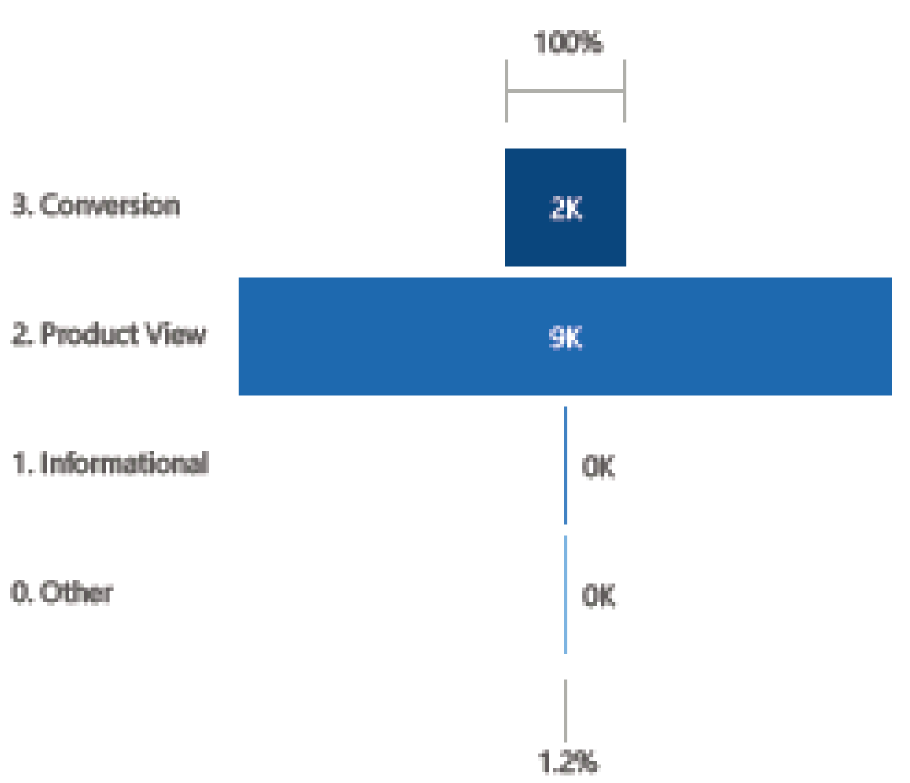
702.31M
avg_page_value

7.15M
avg_bounce_rate

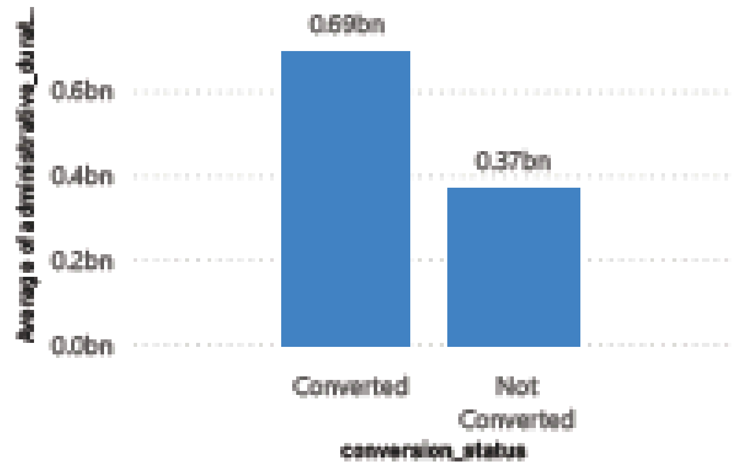
Month
month
2 | 12

Visitor Type
0 | 1 | 2

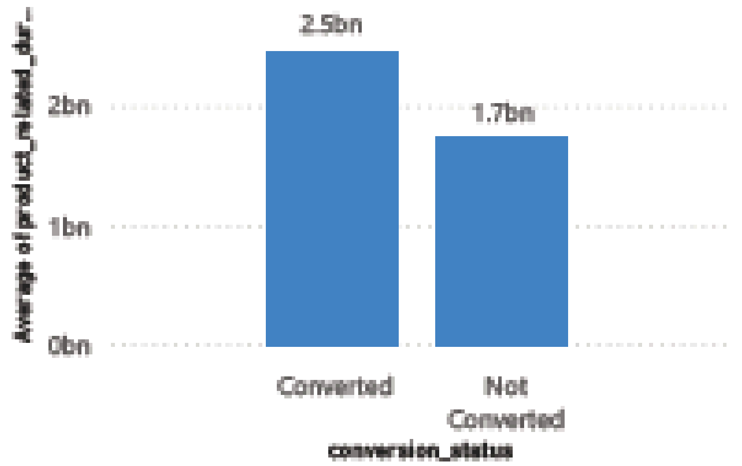
Sessions by Funnel Stage



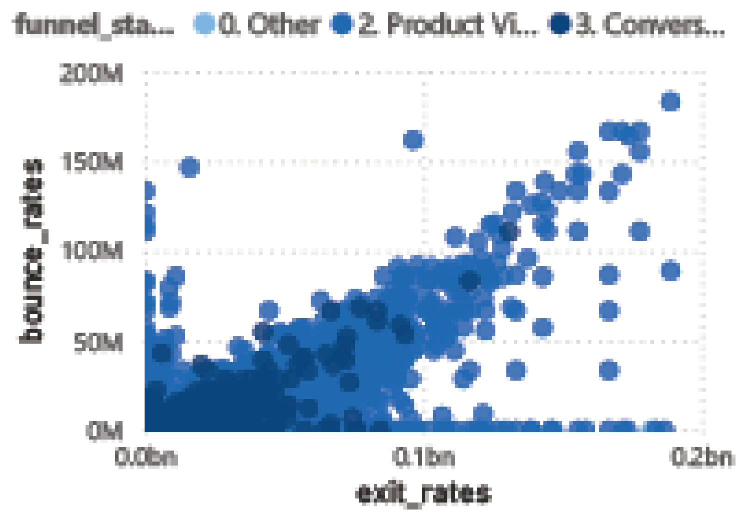
Average of Administrative Duration by Conversion Status



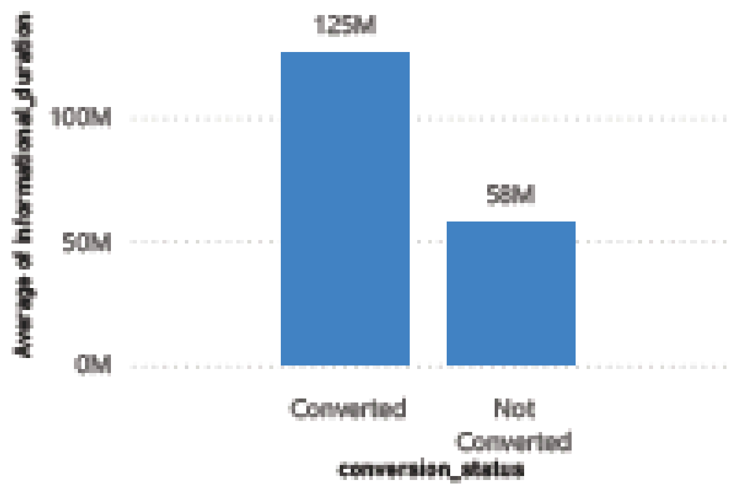
Converters Spend Longer Time on Product Pages



High Bounce & Exit Rates Correlate with Funnel Drop-off



Average of Informational Duration by Conversion Status



SYSTEM SEGMENTATION

0.16	10K	1648	702.31M	7.15M
conversion_rate	total_sessions	total_conversions	avg_page_value	avg_bounce_rate

Region

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8

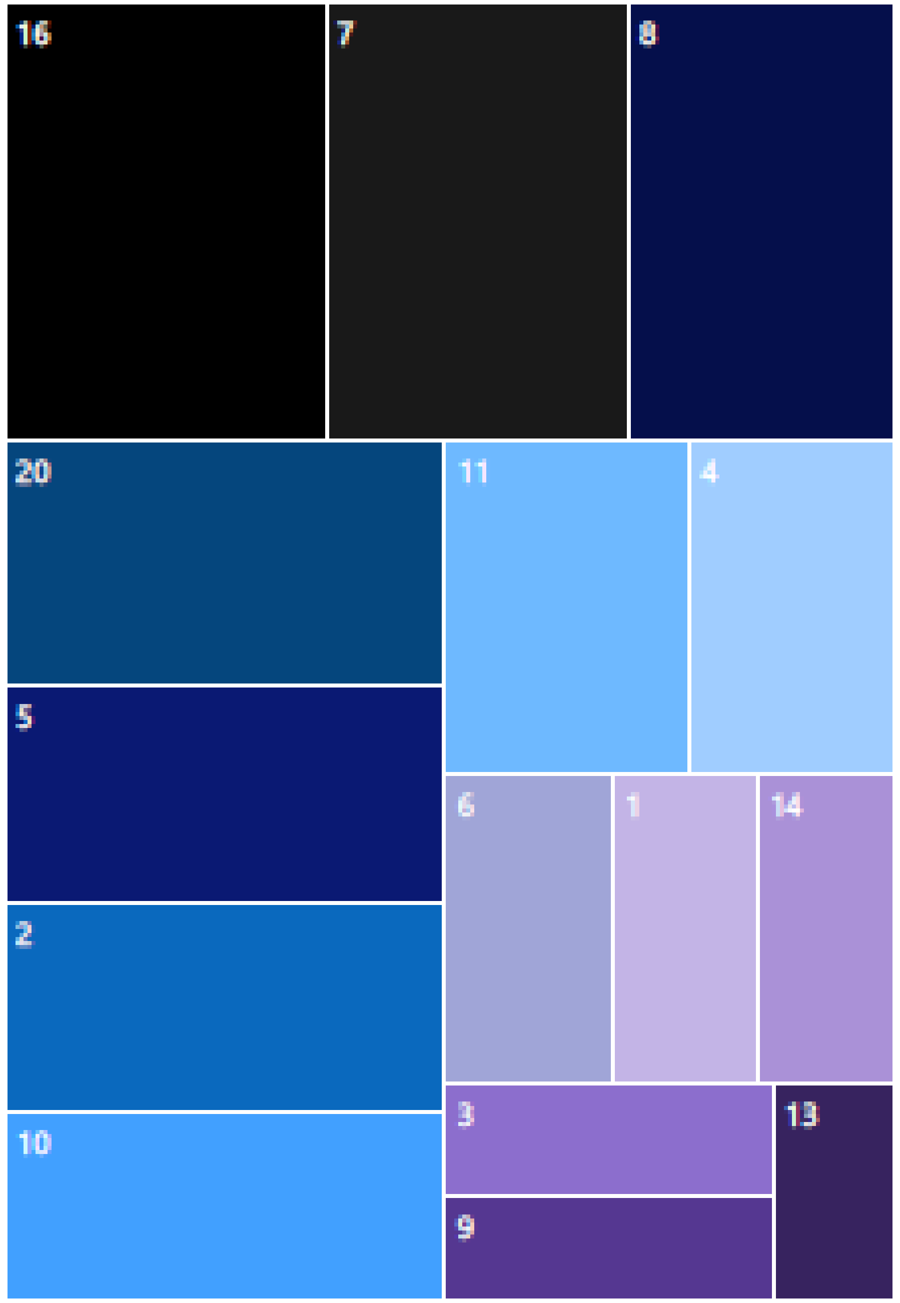
Special Day

- 0
- 2
- 4
- 6
- 8
- 10

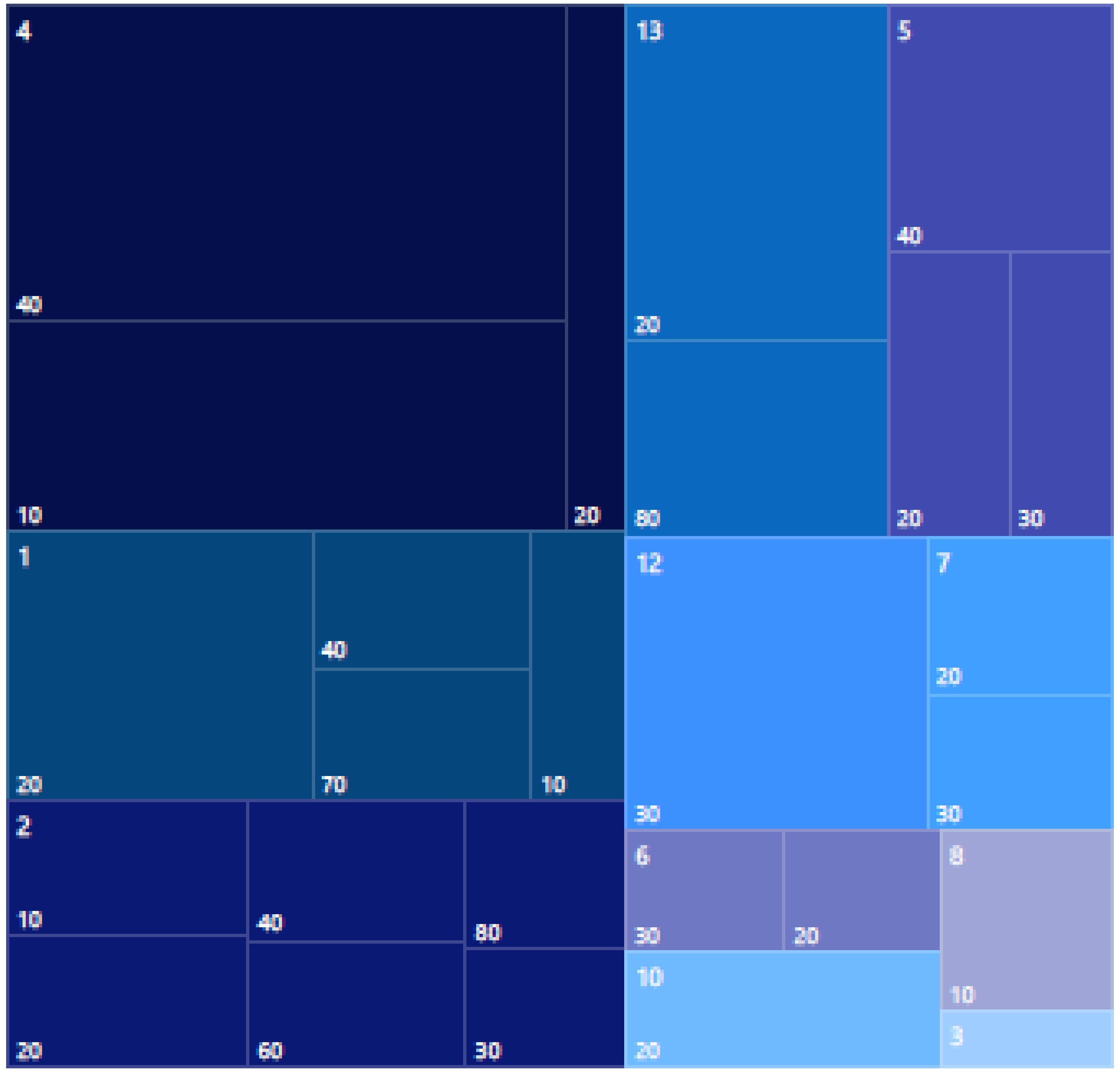
Weekend

0 | 1

Conversion Rate by Traffic Type



Conversion Rate by Browser and Operating Systems



Data Preprocessing

Model 1:

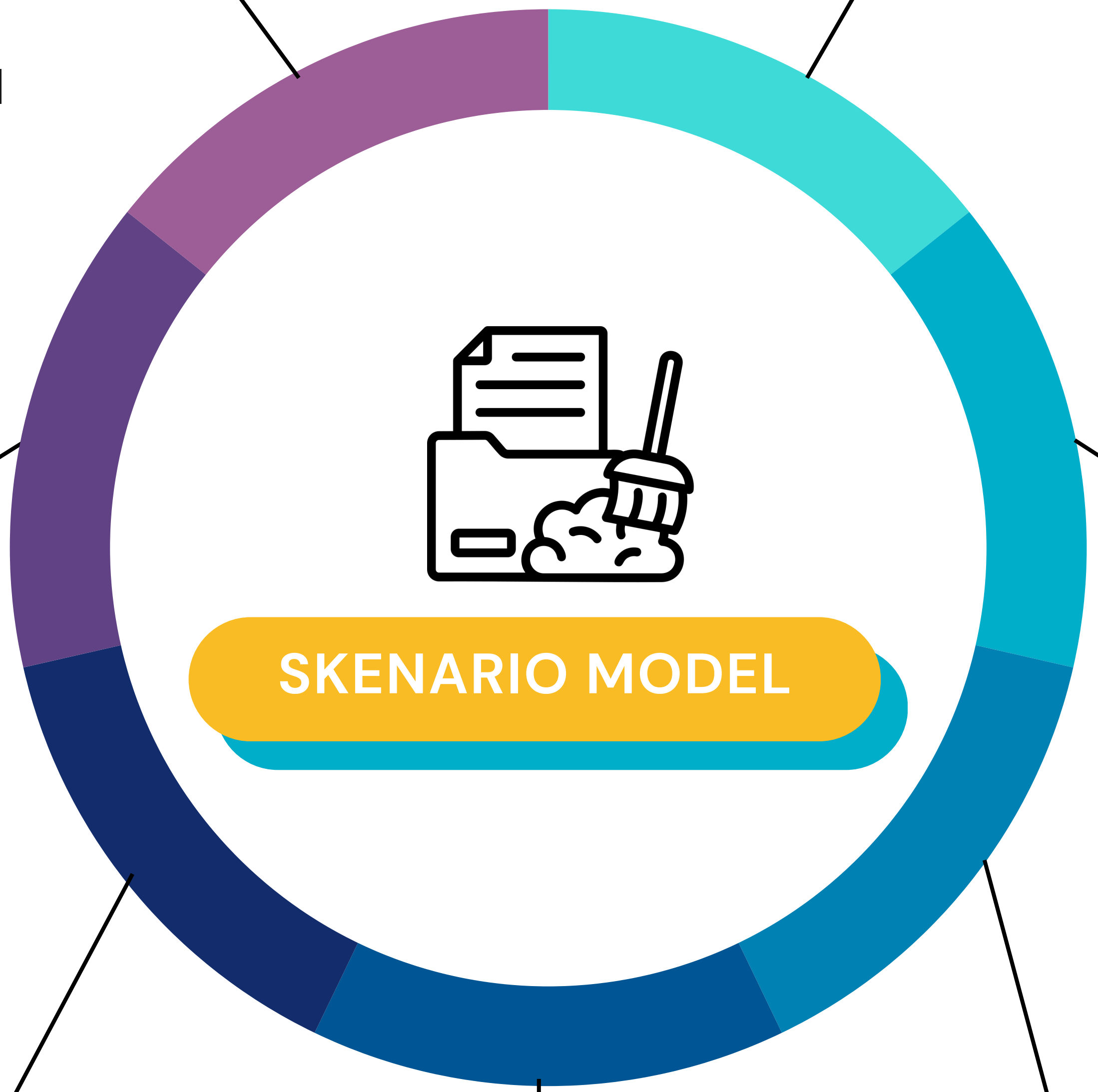
1. Duplicate data was dropped, amounting to **5.5%** of the total data.
2. **All missing values** in all features (administrative, administrative_duration, product_related_duration, and operating_systems) were deleted, with a total deletion percentage of **0.14%** of the duplicate data that had been deleted.

Model 2:

1. Remove duplicates such as **Model 1**.
2. Remove missing values in **administrative** and **bounce_rate** features.
3. Assume missing values in **product_related_duration** and **administrative_duration** with the sum of each (**product_related** or **administrative**) multiplied by 120 seconds.
4. Fill in missing values in **operating_systems** with mode.

Model 3:

Apply treatment as in **Model 2** and remove outlier **exit_rates** of **0.08%** from the data after Model 2 treatment is performed.



Model 4:

Apply the treatment as in **Model 2**. Then, **standardize** the data in x_train and x_test.

Model 5:

Perform treatment as in **Model 2**. Then, perform **feature extraction** on this model by adding the following features.

1. total_visits
2. total_duration
3. visit_bounce_rates
4. visit_exit_rates
5. special_day_visits
6. special_day_duration

Model 6:

Treat it like **Model 3**, then perform oversampling with SMOTE accompanied by several **oversampling** comparisons as follows.

1. Resampled 50:50
2. Resampled 50:40
3. Resampled 50:35
4. Resampled 50:30
5. Resampled 50:25
6. Resampled 50:20

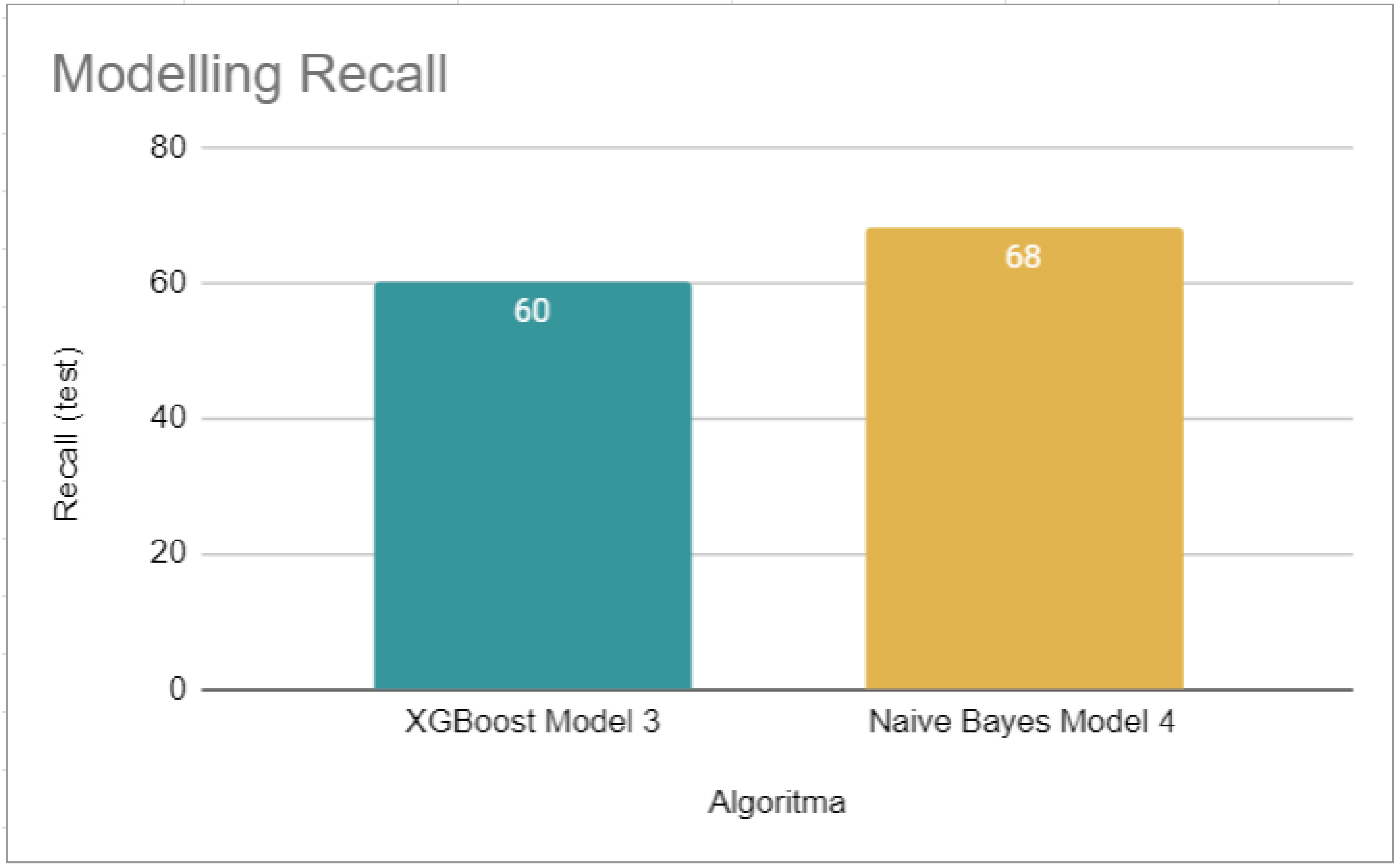
Model 7:

In addition to performing the same treatment as **model 6**, data **standardization** was performed on each resampled x_train and x_test.

Modelling

The metric we focused on evaluating was **Recall**, with the goal of **minimizing False Negatives** so that **we don't lose potential customers**.

Modeling was performed based on preprocessing data to observe the performance of **7 modeling scenarios** so as **not to select overfitted results**.



$$\text{Recall} = \frac{TP}{TP + FN}$$

Of all the modeling scenarios, **modeling 3 with XGBoost** and **modeling 4 with Naive Bayes** have great potential, with performance that is quite good compared to the others. -> **Optimization of these 2 models**

RECALL SETELAH HYPERPARAMETER TUNNING

XG Boost	Naive Bayes
Tuned parameters: max_depth : [3, 5, 7] learning_rate : [0.1, 0.01, 0.001] subsample : [0.5, 0.7, 1]	**Tuned parameters:** priors : [None, [0.4, 0.6], [0.5, 0.5],[0.3, 0.7],[0.2,0.8],[0.1,0.9]] var_smoothing : np.logspace(0, -9, num=100) }
59	97

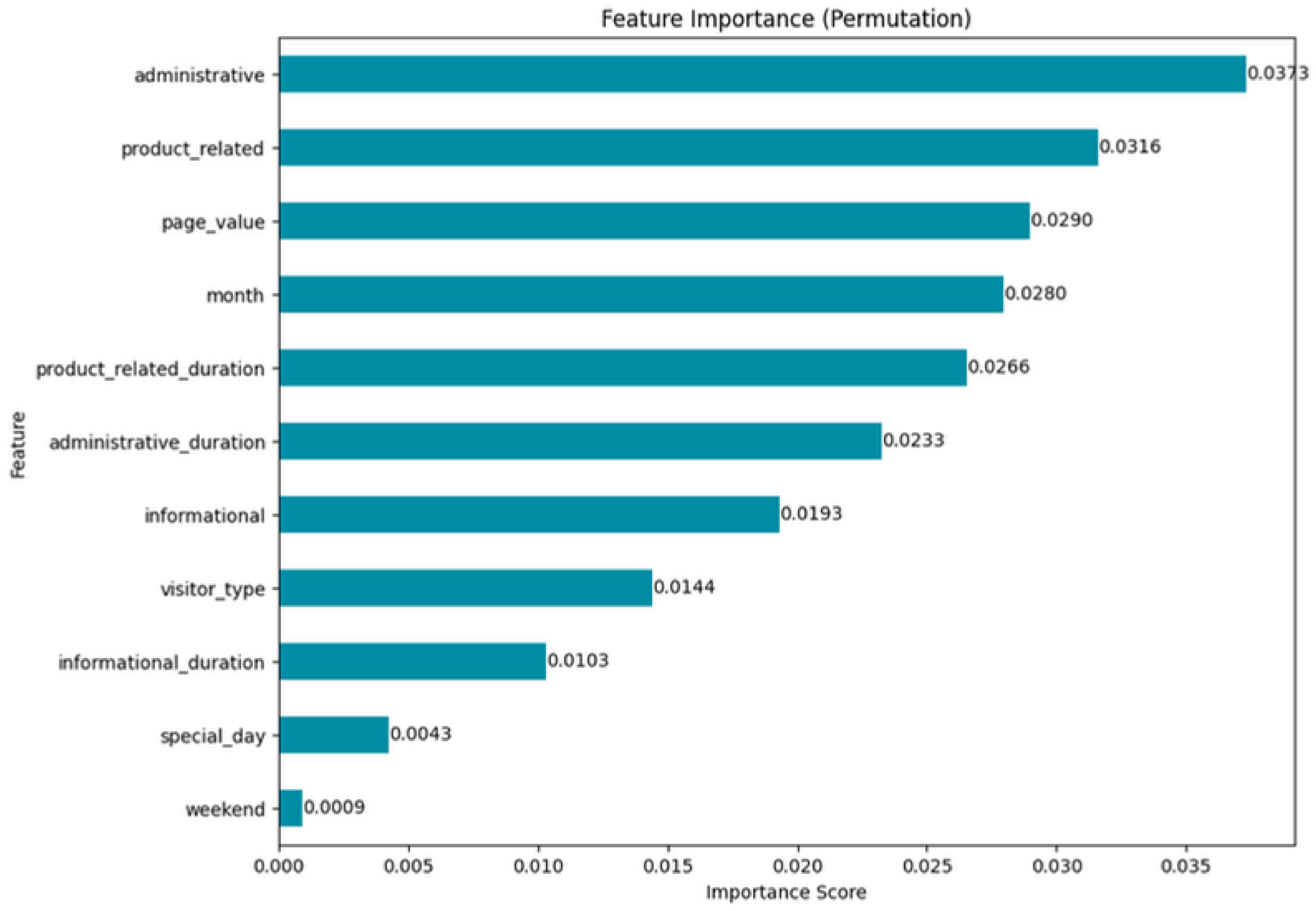
Recall (Test Set): 0.97, indicating that the model is **able to detect 97% of all actual conversion cases**.

With such a high recall, the model is highly reliable in ensuring that **no conversions are missed**, in line with the model's main priority of identifying all conversions.

Confusion Metrix Model Naive Bayes

		Predicted	
		No	Yes
Actual	No	TN 2367	FP 7811
	Yes	FN 36	TP 1847

FEATURE IMPORTANCE



From the tuned Naive Bayes model, it appears that there are **four features that are very important in predicting whether a visitor** will convert into a customer or not, namely:

1. Administrative
2. Product Related
3. Month
4. Page Value
5. Product Related Duration.

These features have the **most significant impact on user conversion rates**, which supports the previous analysis results in EDA (in the correlation heatmap). **Therefore, it is highly recommended that companies focus on optimizing these features.**

Business Impact

BUSINESS IMPACT

Conversion Rates Changes(CVR)

- raw data conversion : **15,59%**
- **conversion rate prediction model** : **81,63%**
- **Improvement:** 81,63% - 15,59% = **66,37%**

Total Conversions vs. Non-Conversions

- **Actual data from the dataset, 12,061 visitors (no duplicates)**
- **Conversion: 81.63%** of 12,061 ≈ 9,845 visitors
- **Non-conversion: 18.37%** of 12,235 ≈ 2,216 visitors

Revenue Assumptions

- **Suppose the average revenue per conversion** is **$50**
- **Before Model Implementatio**n: 1,880 visitors x $50 = **$94,000**
- **After Model Implementation**: 9,845 visitors x $50 = **$492,250**
- **Increase in Visitors**: 9,845 - 1,880 = **7,965 visitors**
- **Increase in Revenue**: $492,250 - $94,000 = **$398,250**

Recommendation

01

Identifying factors that influence customer conversion

FEATURE IMPORTANCE

1. administrative
2. product_related
3. page_value
4. month
5. product_related_duration
6. administrative_duration
7. informational
8. visitor_type
9. informational_duration
10. special_day
11. weekend

It is necessary to **focus on optimizing the top 5 features** to increase the conversion rate.

02

Increasing company sales

Insight

- **Page Value**, **Product Related**, and **Administrative** features **significantly increase** the chances of visitors making a purchase.

- Returning visitors dominate website traffic with low visitor conversion rates, **indicating that some visitors enjoy browsing products**, but the **products they encounter do not match their preferences.**

- **There was an increase** in conversion rates in **May** and **November**, as well as on weekends.

Recommendation

- Focus on increasing **Page Value** and **Product Relatedness** by providing relevant and engaging content to drive conversions.
  - Optimize product descriptions
  - Leverage company data (visitor activity) to implement product recommendations based on each visitor's preferences.

- Take advantage of weekends and the months leading up to May and November (holiday season or year-end) to optimize marketing.

03

Anticipating a further decline in sales

Properly implement prediction models and conduct evaluations and regular monitoring of campaign and model performance.

THANK YOU

APPENDIX

Model	Algoritma	Accuracy (test)	Accuracy (train)	Recall (test)	roc_auc (test)	roc_auc (train)	crossval (train)	crossval (test)	CV	
	XGBoost	90	99	60	92	100	94	56	89	Overfit
	ETC	90	100	50	92	100	100	47	89	
1	Naive Bayes	84	85	53	83	84	53	50	84	
	XGBoost	90	99	61	93	100	92	57	89	Overfit
	ETC	90	100	53	92	100	100	48	89	
2	Naive Bayes	85	84	53	85	83	52	49	84	
	XGBoost	86	99	60	91	100	95	60	87	
	ETC	85	100	47	90	100	100	50	87	
3	Naive Bayes	81	82	40	78	81	44	42	81	
	XGBoost	90	99	61	93	100	95	60	89	
	ETC	87	100	56	74	100	100	56	86	
4	Naive Bayes	82	81	68	84	82	46	43	80	
	XGBoost	89	99	60	93	100	94	56	89	Overfit
	ETC	90	100	46	92	100	100	42	88	
5	Naive Bayes	84	83	59	84	82	58	55	82	
	XGBoost									
	50:50:00	89	99	65	92	100	92	56	91	
	50:40:00	89	99	64	92	100	92	56	91	
	50:35:00	89	99	65	92	100	92	56	91	
	50:30:00	89	99	62	92	100	92	56	91	
	50:20:00	89	99	66	92	100	92	56	90	Overfit
	50:25:00	89	99	62	92	100	92	56	90	
	ETC									
	50:50:00	89	99	65	92	100	92	56	93	
	50:40:00	89	100	67	92	100	100	46	92	
	50:35:00	89	100	63	92	100	100	46	92	
	50:30:00	90	100	64	92	100	100	47	92	
	50:20:00	90	100	63	92	100	100	47	91	
	50:25:00	90	100	62	92	100	100	47	91	Overrfit
	Naive Bayes									
	50:50:00	80	78	66	81	86	52	49	77	
	50:40:00	81	78	65	81	85	52	49	77	
	50:35:00	82	79	63	82	85	52	49	77	
	50:30:00	83	79	62	82	85	52	49	78	
	50:20:00	85	80	58	83	84	52	49	80	
6	50:25:00	84	79	62	82	85	52	49	78	
	XGBoost									
	50:50:00	16	99	100	56	100	92	56	91	
	50:40:00	16	99	100	56	100	92	56	91	
	50:35:00	16	99	100	61	100	92	56	91	
	50:30:00	17	99	99	71	100	92	56	91	
	50:20:00	87	99	76	91	100	92	56	90	
	50:25:00	89	99	62	92	100	92	56	90	
	ETC									
	50:50:00	62	100	89	83	100	100	47	93	
	50:40:00	73	100	85	86	100	100	46	92	
	50:35:00	76	100	81	86	100	100	47	92	
	50:30:00	85	100	71	89	100	100	47	92	
	50:20:00	90	84	59	85	82	92	56	91	
	50:25:00	89	100	63	92	100	100	47	91	
	Naive Bayes									
	50:50:00	48	73	53	50	85	66	64	73	
	50:40:00	51	72	50	50	85	66	64	72	
	50:35:00	53	72	48	50	85	66	64	72	
	50:30:00	54	72	44	50	84	66	64	71	
	50:20:00	60	74	35	50	84	66	64	73	
7	50:25:00	56	72	40	50	84	66	64	72	

The results of the modeling experiment using the 3 best models show higher recall values but tend to be overfitted.